

Data Communication & Computer Networks (CS24305) — Full Course Master Book

Complete End-to-End B.Tech CSE 5th Semester Study Book & PYQ Bank

EXECUTIVE MASTER STUDY GUIDE & PROTOCOL BANK

Data Communication & Computer Networks (CS24305)

Birla Institute of Technology, Mesra | B.Tech CSE 5th Semester (NEP 2024–25 Scheme)

Complete Course Structure & Protocol Matrix

Module	Layer & Focus	Key Formulations, RFCs & Protocols
Module I	Physical Layer & Signals	Nyquist & Shannon Capacity, Transmission Impairments, Line Coding (Manchester, AMI), Multiplexing (FDM, TDM, WDM)
Module II	Data Link Layer & MAC	Framing, CRC-32 Polynomial Derivations, Sliding Window ARQs (Go-Back-N, Selective Repeat), CSMA/CD & CSMA/CA
Module III	Network Layer & Routing	IPv4/IPv6 Header Analysis, CIDR Subnetting Calculations, Dijkstra Shortest Path, Bellman-Ford, OSPF, BGP-4
Module IV	Transport Layer Protocols	TCP 3-Way Handshake, TCP State Machine, AIMD Congestion Control, Slow Start, Fast Retransmit, UDP Sockets
Module V	Application Layer & Security	DNS Hierarchical Resolution, HTTP/1.1 vs HTTP/2/3, TLS 1.3 Handshake, RSA Asymmetric Encryption, AES

Exam Preparation & High-Yield Strategy

This publication-grade master book consolidates all 5 modules with formal mathematical channel capacity proofs, step-by-step worked subnetting & CRC numericals, packet header diagrams, and model answers to BIT Mesra end-semester examination questions.

Module I: Data Communication & Networking Overview — Complete 13-Topic Tracker

Topic 1: Communication Model (Sender, Message, Medium, Protocol)

Topic 3: Networks (PAN, LAN, MAN, WAN Taxonomies)

Topic 5: OSI 7-Layer Reference Model & Layer Functions

Topic 7: Standards & Protocol Layers (Modularity & Encapsulation)

Topic 9: Data Transmission Concepts (Bit vs. Baud, Latency = $T_t + T_p + T_q + T_{pr}$)

Topic 11: Digital Data Transmission (Discrete Levels vs. Analog)

Topic 13: Channel Capacity (Nyquist Noiseless vs. Shannon Noisy Capacity)

Topic 2: Data Communications (Simplex, Half-Duplex, Full-Duplex)

Topic 4: The Internet (Packet Switching & TCP/IP Suite)

Topic 6: TCP/IP 4-Layer Protocol Architecture (OSI vs. TCP/IP)

Topic 8: Internet Applications (HTTP, SMTP, DNS, FTP, DHCP)

Topic 10: Analog Data Transmission (Amplitude, Frequency, Phase)

Topic 12: Transmission Impairments (Attenuation, Distortion, Noise)

Topic 1 & 2: Communication Model & Transmission Modes

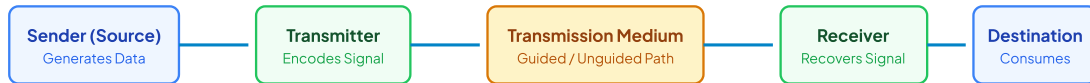


Figure 1.1: Simplified Communication Model & Five Essential Components (S-M-M-P)

Memory Hook: S-M-M-P

Sender → Message → Medium → Protocol (Syntax, Semantics, Timing)

Data Transmission Modes [UPLOADED PYQ]:

Mode	Directional Flow & Channel Mechanics	Real-World Example
1. Simplex	Unidirectional communication ($A \rightarrow B$). Only one device transmits; the other only receives. Channel capacity is 100% dedicated to one direction.	Keyboard to CPU, Television/Radio broadcast.
2. Half-Duplex	Bidirectional communication ($A \rightarrow B$ or $A \leftarrow B$), but not simultaneously . Both can transmit, but must alternate turns.	Walkie-Talkie systems.
3. Full-Duplex	Simultaneous bidirectional transmission ($A \rightleftarrows B$). Both stations can transmit and receive concurrently by dividing bandwidth.	Telephone networks, modern switched Ethernet.

Topic 3 & 4: Network Types & The Global Internet

Network	Geographic Coverage	Ownership & Speed	Representative Example
PAN	≈ 10 meters	Private; moderate speed (Bluetooth / Zigbee).	Phone ↔ Smartwatch ↔ Earbuds
LAN	Single room, office, or campus (≤ 1 km).	Privately owned; high data rate (1 Gbps–10 Gbps).	University campus / Lab LAN
MAN	City or metropolitan area (5–50 km).	Public/Private consortium; high speed.	Cable TV network across a city
WAN	Country, continent, or global (> 100 km).	Multiple telecom operators; variable speed.	The Global Internet

Topic 5 & 6: OSI 7-Layer Model vs. TCP/IP Architecture [UPLOADED PYQ]

OSI Layer	TCP/IP Layer	PDU Name	Core Functional Responsibility
7. Application	1. Application Layer	Data / Message	Provides network services to user apps (HTTP, SMTP, DNS, FTP).
6. Presentation			Syntax translation, data encryption/decryption (TLS), data compression.
5. Session			Dialog control, session checkpointing, synchronization tokens.
4. Transport	2. Transport Layer	Segment (TCP) / Datagram (UDP)	Process-to-process delivery, port addressing, flow control, reliability.
3. Network	3. Internet Layer	Packet / Datagram	Host-to-host routing, logical IP addressing, packet forwarding (IP, ICMP).
2. Data Link	4. Network Access Layer	Frame	Hop-to-hop framing, physical MAC addressing, CRC error detection, MAC access.
1. Physical		Raw Bits ('0'/'1')	Raw bit transmission over physical media, voltage levels, connectors.

🧠 Memory Trick for OSI 7 Layers

All People Seem To Need Data Processing

(Application → Presentation → Session → Transport → Network → Data Link → Physical)

Topic 7 & 8: Standards, Encapsulation & Internet Applications

Encapsulation occurs when data travels down the protocol stack at the sender: each layer prepends a protocol header containing control metadata:

Data $\xrightarrow{\text{Transport}}$ TCP Header + Data (Segment) $\xrightarrow{\text{Network}}$ IP Header + Segment (Packet) $\xrightarrow{\text{Data Link}}$ Frame Header + Packet + CRC Trailer (Frame)

Topic 9 – 11: Data Transmission Concepts, Latency & Digital Signals

1. Total End-to-End Latency Formula:

$$\text{Latency} = T_{\text{trans}} + T_{\text{prop}} + T_{\text{proc}} + T_{\text{queue}}$$

- **Transmission Delay** (T_{trans}): Time to push packet bits onto physical link: $T_{\text{trans}} = \frac{L \text{ (bits)}}{B \text{ (bps)}}$. - **Propagation Delay** (T_{prop}): Time for a bit to physically travel across distance: $T_{\text{prop}} = \frac{d \text{ (meters)}}{v \text{ (speed of light in medium)}}$. - **Bit Rate vs. Baud Rate**: Bit Rate = Baud Rate $\times \log_2(M)$ bps, where M is number of distinct signal voltage levels.

Topic 12: Transmission Impairments [UPLOADED PYQ]

Impairment	Physical Mechanism & Impact	Engineering Mitigation
1. Attenuation	Signal energy loss over distance due to medium resistance. Signal power drops logarithmically (dB).	Periodic Amplifiers (analog) or Regenerative Repeaters (digital).
2. Distortion	Signal changes shape because different frequency harmonic components travel at different phase velocities.	Equalizers and phase delay compensators.
3. Noise	Unwanted extraneous electrical signals injected into the channel: • <i>Thermal Noise</i> : Agitation of electrons ($N = kTB$). • <i>Crosstalk</i> : Electromagnetic coupling between adjacent wire pairs. • <i>Impulse Noise</i> : Spikes from lightning, power lines.	Shielding (STP/Coax), differential twisted pairs, optical fiber.

Topic 13: Channel Capacity Theorems [UPLOADED PYQ]

1. Nyquist Bit Rate (For Noiseless Channels):

$$C = 2 \times B \times \log_2(L) \text{ bps}$$

Where B is channel bandwidth in Hertz (Hz) and L is number of discrete signal voltage levels.

2. Shannon Channel Capacity (For Noisy Channels with Thermal Noise):

$$C = B \times \log_2(1 + \text{SNR}) \text{ bps}$$

Where $\text{SNR} = 10^{\frac{\text{SNR}_{\text{dB}}}{10}}$ is the linear Signal-to-Noise Ratio.

 [UPLOADED PYQ] Numerical: Shannon Capacity for Bandwidth 4 kHz and $\text{SNR}_{\text{dB}} = 24 \text{ dB}$

1. Linear $\text{SNR} = 10^{\frac{24}{10}} = 10^{2.4} \approx \mathbf{251.19}$.
2. $C = 4000 \times \log_2(1 + 251.19) = 4000 \times \log_2(252.19) \approx 4000 \times 7.978 = \mathbf{31,913 \text{ bps}}$ ($\approx 32 \text{ kbps}$).

 M1 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ] Compare OSI and TCP/IP models across 5 engineering parameters. (8 Marks)

1. **Layer Count:** OSI has 7 layers; TCP/IP has 4 (or 5) layers.
2. **Origins:** OSI was designed by ISO as a formal conceptual reference model before protocols were written; TCP/IP was designed pragmatically around actual working protocols (ARPANET).
3. **Services vs Protocols:** OSI strictly separates Services, Interfaces, and Protocols; TCP/IP combines services into protocols directly.
4. **Network Layer Communication:** OSI supports both Connectionless and Connection-Oriented at Network layer; TCP/IP supports only Connectionless (IP).
5. **Transport Layer Communication:** OSI supports only Connection-Oriented; TCP/IP supports both Connection-Oriented (TCP) and Connectionless (UDP).

Module II: Transmission Media & Signal Encoding — Complete 6-Topic Tracker

Topic 14: Guided Media (Twisted Pair UTP/STP, Coaxial Cable, Optical Fiber)

Topic 16: Digital Signaling (Line Codes: NRZ-L, NRZ-I, Manchester, AMI)

Topic 18: Encoding Techniques (Line Coding Comparison & Tradeoffs)

Topic 15: Wireless Media & Propagation (Ground, Sky, LOS, Satellite Microwave)

Topic 17: Analog Signaling (Carrier Parameter Representations)

Topic 19: Modulation Techniques (ASK, FSK, PSK & Constellation QAM)

Topic 14: Guided Transmission Media

Guided Medium	Physical Architecture & Transmission Physics	Key Advantages & Tradeoffs
1. Twisted Pair Cable	Two insulated copper wires twisted in a helical spiral. Twisting ensures both wires receive equal electromagnetic interference (EMI), which cancels out at the differential receiver. Available as UTP (Unshielded) and STP (Shielded with metal foil).	Inexpensive, lightweight, easy to terminate with RJ-45 connectors. Limited bandwidth (≤ 10 Gbps over short distances), prone to crosstalk.
2. Coaxial Cable	Solid central copper conductor surrounded by dielectric insulating layer, enclosed in a braided cylindrical wire mesh shield and outer protective plastic jacket.	Better noise immunity and higher bandwidth than twisted pair. Bulkier, stiffer, harder to install. Used in Cable TV broadband (DOCSIS).
3. Optical Fiber	Ultra-pure cylindrical glass core (n_1) surrounded by concentric glass cladding (n_2) where refractive index $n_1 > n_2$. Transmits light pulses via Total Internal Reflection whenever incident angle $\theta > \theta_{\text{critical}}$.	Enormous bandwidth (terabits/sec), ultra-low attenuation (≈ 0.2 dB/km), zero EMI susceptibility, lightweight, highly secure against wiretapping. Expensive splicing and optical transceivers.

Topic 15: Wireless Transmission & Propagation [UPLOADED PYQ]

Propagation Mode	Atmospheric Interaction & Frequency Band	Representative Applications
1. Ground-Wave Propagation	Signals follow the curvature of the earth via diffraction over the conductive ground surface (< 2 MHz).	AM radio broadcasting, maritime navigation.
2. Sky-Wave Propagation	Signals bounce back and forth between earth's surface and the charged ionosphere (2–30 MHz).	Shortwave (SW) international radio broadcasting, amateur radio.
3. Line-of-Sight (LOS)	Straight-line path between transmitting and receiving directional horn antennas (> 30 MHz). Antennas must have clear Fresnel zone.	Terrestrial microwave links, cellular base stations, Wi-Fi.

Satellite Microwave vs. Terrestrial Microwave [UPLOADED PYQ]:

Terrestrial Microwave: Uses parabolic dish antennas mounted on high towers spaced 30–50 km apart due to Earth's horizon curvature. Low propagation delay (< 1 ms).

Satellite Microwave: Earth station beams signal to a geostationary satellite at 35,786 km altitude (Uplink), which amplifies and retransmits it on a different downlink frequency. High round-trip latency (≈ 270 –540 ms), but provides massive continental coverage.

Topic 16 & 18: Digital Line Coding Schemes [UPLOADED PYQ]

[UPLOADED PYQ] Step-by-Step Waveform Rules for Bit Stream '001110011'

Encoding Scheme	Bit Transition Invariant Rule	Waveform Encoding for '001110011'
NRZ-L (Level)	Bit '0' = Positive voltage ($+V$), Bit '1' = Negative voltage ($-V$).	'+V, +V, -V, -V, -V, +V, +V, -V, -V'
NRZ-I (Invert)	Bit '1' = Transition at start of bit interval; Bit '0' = No transition.	Maintains previous level on '0', flips on '1'.
Bipolar AMI	Bit '0' = Zero voltage ($0V$); Bit '1' = Alternating $+V$ and $-V$.	'0V, 0V, +V, -V, +V, 0V, 0V, -V, +V'
Manchester (802.3)	Bit '0' = High-to-Low transition at bit center; Bit '1' = Low-to-High transition at center.	Guarantees a mid-bit clock transition in every single bit!
Differential Manchester	Always transitions at center. Bit '0' = Transition at start; Bit '1' = No transition at start.	Token Ring standard; immune to wire polarity inversion.

Topic 17 & 19: Carrier Modulation Techniques (ASK, FSK, PSK, QAM)

Digital modulation alters one or more parameters of a high-frequency analog sinusoidal carrier $c(t) = A \cos(2\pi f_c t + \phi)$:

Modulation	Modulated Parameter	Mathematical Signal Equation	Key Advantages & Tradeoffs
ASK	Amplitude	$s(t) = A_1 \cos(2\pi f_c t)$ for '1', $A_0 \cos(2\pi f_c t)$ for '0'	Simple; highly susceptible to noise and fading.
FSK	Frequency	$s(t) = A \cos(2\pi f_1 t)$ for '1', $A \cos(2\pi f_2 t)$ for '0'	High noise immunity; requires more channel bandwidth.
BPSK / QPSK	Phase	$s(t) = A \cos(2\pi f_c t + \phi_i)$ where $\phi_i \in \{0, \pi\}$	Excellent bit error rate (BER); standard in modern wireless.
QAM (16/64/256)	Amplitude + Phase	$s(t) = I(t) \cos(2\pi f_c t) - Q(t) \sin(2\pi f_c t)$	Enormous spectral efficiency ($\log_2 M$ bits/ baud); used in Wi-Fi & 5G.

M2 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ] Draw the line coding waveforms for bit stream '001110011' using NRZ-L, NRZ-I, Bipolar AMI, Manchester, and Differential Manchester. (10 Marks)

- NRZ-L:** High for '0', Low for '1' → 'High, High, Low, Low, Low, High, High, Low, Low'.
- NRZ-I:** Invert signal level on every '1'; no change on '0'.
- Bipolar AMI:** Zero volts for all '0's; alternating $+V$ and $-V$ for each successive '1' → '0, 0, +V, -V, +V, 0, 0, -V, +V'.
- Manchester:** Every bit has mid-interval transition. '0' = High-to-Low; '1' = Low-to-High.
- Differential Manchester:** Mandatory mid-bit transition. Start-of-bit transition for '0'; no start transition for '1'.

Module III: Error Handling, Data Link Control & Multiplexing — Complete Tracker

Topic 20: Types of Errors (Single-Bit Errors vs. Burst Errors)

Topic 22: Checksum (1's Complement Summation & Verification)

Topic 24: Error Correction (Hamming Code & Syndrome Decoding)

Topic 27: High-Level Data Link Control (HDLC Frame: I, S, U-Frames)

Topic 21: Error Detection Techniques (Simple Parity & Limitations)

Topic 23: Cyclic Redundancy Check (CRC-32 Modulo-2 Division)

Topic 25 & 26: Flow & Error Control ARQ (Stop-and-Wait, GBN, Selective Repeat)

Topic 28 & 29: Multiplexing (FDM, Synchronous TDM vs. Statistical TDM)

Topic 20 – 22: Error Types & Basic Detection Techniques

Error Class	Mathematical Definition	Detection Feasibility
1. Single-Bit Error	Only 1 bit in the data block is inverted from 0 $\rightarrow$ 1 or 1 $\rightarrow$ 0.	Easily detected by a single Parity Bit.
2. Burst Error	Two or more bits in a sequence are corrupted. Length of burst is measured from first corrupted bit to last corrupted bit.	Requires Cyclic Redundancy Check (CRC) or 2D Parity.

Topic 23: Cyclic Redundancy Check (CRC) Polynomial Division [UPLOADED PYQ]

 **[UPLOADED PYQ] Solved Problem: CRC for Data `1010011110` with Generator $G = 1011$**

Step 1: Determine Degree of Generator: $G = 1011$ has degree $r = 3$.

Step 2: Append $r = 3$ zeros to Data: Augmented Dividend = `1010011110000`.

Step 3: Modulo-2 XOR Division:

```
      1001001011 (Quotient)
1011 ) 1010011110000
      1011
      ---
      001011
       1011
       ---
       00001110
        1011
        ---
        1010
        1011
        ---
        001000
         1011
         ---
         0110  $\leftarrow$  Remainder CRC = 011
```

Step 4: Transmitted Frame: Data + CRC = **1010011110011**.

Topic 24: Error Correction (Hamming Code (n, k) Derivation)

Hamming Parity Bit Invariants (Hamming, 1950): For m data bits and r parity check bits:

$$2^r \geq m + r + 1$$

- Parity bits are placed strictly at bit positions that are powers of 2 (1, 2, 4, 8, 16, ...). - Data bits occupy all remaining positions (3, 5, 6, 7, 9, 10, ...). - Parity bit P_k checks all bit positions whose binary representation has a 1 in the k -th position. - **Syndrome:** Evaluation of parity checks at the receiver forms a binary integer: 000 $\Rightarrow$ No error; non-zero value indicates the exact corrupted bit position!

Topic 25 & 26: Sliding Window Flow & Error Control ARQ Protocols [UPLOADED PYQ]

ARQ Protocol	Sender Window (W_s)	Receiver Window (W_r)	Key Retransmission Behavior & Efficiency
1. Stop-and-Wait ARQ	$W_s = 1$	$W_r = 1$	Sender waits for ACK before sending next frame. Low efficiency on long propagation delays ($\eta = \frac{1}{1+2a}$).
2. Go-Back-N (GBN) [UPLOADED PYQ]	$W_s \leq 2^m - 1$	$W_r = 1$ (Cumulative ACK)	When frame k is lost or damaged, receiver discards all subsequent out-of-order frames. Sender retransmits all frames from k onwards.
3. Selective Repeat (SR)	$W_s \leq 2^{m-1}$	$W_r \leq 2^{m-1}$ (Independent ACK)	Receiver buffers out-of-order frames. Sender retransmits <i>only the specific missing frame k</i> , saving bandwidth.

Topic 27: High-Level Data Link Control (HDLC Frame Formats)

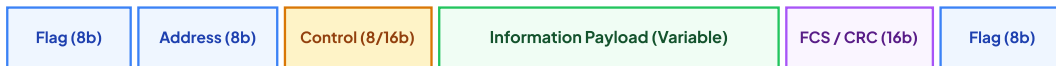


Figure 3.1: Universal HDLC Frame Architecture with Flag '01111110' Delimiters

The 3 HDLC Frame Formats

- 1. **Information Frames (I-Frames):** Carries user data payload and piggybacked ACKs ($N(S)$ send sequence and $N(R)$ receive sequence).
- 2. **Supervisory Frames (S-Frames):** Carries flow/error control without payload (RR - Receive Ready, RNR - Receive Not Ready, REJ - Reject / NAK).
- 3. **Unnumbered Frames (U-Frames):** Used for link initialization, mode setting (SABM - Set Asynchronous Balanced Mode), and link disconnection (DISC).

Topic 28 & 29: Multiplexing (FDM, Synchronous vs. Statistical TDM) [UPLOADED PYQ]

Feature	Synchronous TDM	Statistical / Asynchronous TDM [UPLOADED PYQ]
Time Slot Allocation	Pre-assigned, fixed, dedicated time slots to each input line in a round-robin cycle.	Time slots allocated dynamically on-demand only to active input lines with data to send.
Bandwidth Efficiency	Low: if an input line has no data to transmit, its allocated time slot goes completely empty/wasted.	High: zero wasted empty slots. Total input data rate can exceed the aggregate multiplexed link capacity.
Addressing Overhead	No addressing needed in slot (receiver knows source by slot position).	Each slot requires header addressing metadata to identify source/destination.

M3 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ] Differentiate between Stop-and-Wait ARQ and Go-Back-N ARQ. (8 Marks)

1. **Sender Window Size:** Stop-and-Wait has $W_s = 1$; Go-Back-N has $W_s = 2^m - 1 > 1$, enabling continuous transmission pipelining.
2. **Channel Utilization:** Stop-and-Wait has very poor link utilization ($\eta = \frac{1}{1+2a}$); GBN achieves near 100% utilization when $W_s \geq 1 + 2a$.
3. **Retransmission Scope:** In Stop-and-Wait, only 1 frame is retransmitted on timeout; in GBN, the entire window of N unacknowledged frames is retransmitted upon a single lost frame.

Module IV: Wide Area Networks (WAN) & Local Area Networks (LAN) — Complete Tracker

Topic 30: Switching Networks (Circuit vs. Packet Switching Paradigms)

Topic 32: Circuit-Switching Concepts (Blocking, Dedicated Resources)

Topic 34: Cellular Network Principles (Cell Geometry, Frequency Reuse, Handoff)

Topic 36: Network Topologies (Bus, Star, Ring, Mesh, Tree Analysis)

Topic 38: Virtual Local Area Networks (VLAN Security & Broadcast Domains)

Topic 31: Circuit-Switching Networks (3 Phases: Setup, Transfer, Teardown)

Topic 33: Packet-Switching Principles (Datagram vs. Virtual-Circuit)

Topic 35: Cellular Generations Evolution (1G Voice to 5G Ultra-Low Latency)

Topic 37: LAN Protocol Architecture (IEEE 802: LLC vs. MAC Sublayers)

Topic 30 – 33: Switching Paradigms (Circuit vs. Packet Switching) [UPLOADED PYQ]

Parameter	Circuit Switching	Packet Switching [UPLOADED PYQ]
Path Establishment	Dedicated physical path established before data transfer (Setup → Transfer → Teardown).	No dedicated end-to-end path; data divided into independent packets routed dynamically.
Resource Reservation	Dedicated channel bandwidth reserved 100% (resources wasted if line is idle).	Statistical multiplexing; bandwidth shared on-demand among all active users.
Traffic Suitability	Ideal for continuous, real-time audio/voice communication.	Ideal for bursty, asynchronous computer data traffic (Internet).
Delay Characteristics	Initial connection setup delay; zero queuing delay during active transfer.	Zero setup delay (in datagram); packets experience variable queuing delay at routers.

Datagram vs. Virtual-Circuit Packet Switching [UPLOADED PYQ]:

Datagram Network (Connectionless — IP): Each packet contains full source and destination IP addresses and is treated as an independent entity. Intermediate routers make routing decisions per packet; packets may arrive out of order.

Virtual-Circuit Network (Connection-Oriented — ATM / X.25): A logical circuit is established upfront. Packets carry a short Virtual Circuit Identifier (VCI) rather than full global addresses. All packets follow the exact same route in sequence.

Topic 34 & 35: Cellular Network Principles & Generations (1G – 5G)

Core Cellular Principles: Frequency Reuse & Handoff

- Frequency Reuse:** Geographical service area is divided into regular hexagonal **cells**. Adjacent cells use different frequency sets, but non-adjacent cells separated by minimum reuse distance $D = R\sqrt{3N}$ reuse the same frequencies, dramatically multiplying network capacity!
- Handoff / Handover:** The automatic transition of an active ongoing call/data session from one base station channel to another as the mobile user moves across cell boundaries (Hard Handoff: "Break before make"; Soft Handoff: "Make before break").

Generation	Key Technological Milestone	Core Services & Data Rate	Multiple Access Tech
1G (1980s)	Analog cellular technology.	Voice only (≈ 2.4 kbps).	FDMA
2G (1990s)	Digital cellular, encryption, roaming.	Voice, SMS text (≈ 64 kbps).	TDMA / CDMA (GSM)
3G (2000s)	Mobile broadband Internet (UMTS).	Video calling, mobile web (≈ 2 Mbps).	WCDMA
4G (2010s)	All-IP network, LTE standard.	HD video streaming, gaming (≈ 100 Mbps).	OFDMA
5G (2020s)	Millimeter wave, network slicing, massive IoT.	eMBB, URLLC, mMTC (1–10 Gbps, < 1 ms).	Scalable OFDMA

Topic 36: Physical Network Topologies

Topology	Structural Layout & Communication	Key Advantages & Vulnerabilities
1. Mesh	Every node has dedicated point-to-point links to all other $N - 1$ nodes (Total links = $\frac{N(N-1)}{2}$).	Ultimate fault tolerance and privacy. High cable cost and complex installation.
2. Star	Each node connected to a central controller / switch via dedicated point-to-point link.	Easy to install, isolate faults. Central switch failure halts entire network.
3. Bus	All devices share a single long central backbone cable terminated at both ends.	Low cabling cost. Backbone cable break disables entire network.
4. Ring	Each device has dedicated point-to-point connection to exactly two neighboring nodes in a loop.	Predictable token traffic. Break in ring disables network (unless dual-ring).

Topic 37 & 38: LAN Architecture & Virtual LANs (VLAN)

IEEE 802 LAN Architecture divides the OSI Data Link Layer into two distinct sublayers:

Logical Link Control (LLC — IEEE 802.2): Independent of physical medium; handles flow control, error control, and upper-layer protocol multiplexing.

Medium Access Control (MAC): Dependent on physical topology; governs shared-channel access rules (CSMA/CD in 802.3, CSMA/CA in 802.11) and framing.

VLAN (Virtual Local Area Network — IEEE 802.1Q)

A **VLAN** is a logical broadcast domain created by software configuration on switches regardless of physical cable connections. It isolates network broadcast storms, enhances department security, and reduces routing costs!

M4 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ] Compare Circuit Switching and Packet Switching across 4 architectural parameters. (8 Marks)

1. **Connection Setup:** Circuit switching requires explicit 3-phase connection setup; Packet switching (datagram) sends packets immediately with zero setup.
2. **Resource Utilization:** Circuit switching reserves 100% dedicated channel capacity; Packet switching uses statistical multiplexing to maximize bandwidth utilization.
3. **Congestion & Delays:** Circuit switching has fixed transmission delay once connected; Packet switching suffers from variable queuing and router processing delays.
4. **Failure Resilience:** Circuit switching loses the call if any link on the dedicated path breaks; Packet switching automatically reroutes individual packets around failed links.

Module V: Ethernet, IP, Routing & Applications — Complete 16-Topic Tracker

Topic 39: Traditional Ethernet (IEEE 802.3 CSMA/CD at 10 Mbps)
Topic 41: IEEE 802.11 Wi-Fi (CSMA/CA with RTS/CTS, Ethernet vs. Wi-Fi)
Topic 43: IP Addressing & Subnetting (2^{borrowed} & $2^{\text{host}} - 2$)
Topic 45: Routing in Packet Networks (Hop Count, Cost, Bandwidth)
Topic 47: Link State Routing (Dijkstra's Algorithm & OSPF Flooding)
Topic 49: Congestion Control (Leaky Bucket vs. Token Bucket Shaping)
Topic 51: Simple Mail Transfer Protocol (SMTP, POP3, IMAP)
Topic 53: Hypertext Transfer Protocol (HTTP/1.1 vs. HTTP/2 & Statuses)

Topic 40: High-Speed Ethernet (Fast 100M, Gigabit 1G, 10-Gigabit)
Topic 42: Internet Protocol (IPv4 20-Byte Header vs. IPv6 128-Bit)
Topic 44: Transport Protocols (TCP 3-Way Handshake vs. UDP Sockets)
Topic 46: Distance Vector Routing (Bellman-Ford & RIP Count-to-Infinity)
Topic 48: Path Vector Routing (Border Gateway Protocol BGP-4)
Topic 50: Traffic Management & Quality of Service (QoS Dimensions)
Topic 52: Domain Name System (DNS Hierarchy & Records A, MX)
Topic 54: Dynamic Host Configuration Protocol (DHCP DORA Sequence)

Topic 39 – 41: Ethernet Evolution & Wireless LAN (IEEE 802.11)

Standard	Data Rate	Physical Media	Medium Access Control (MAC)
Traditional Ethernet	10 Mbps	10Base5 (Thick coax), 10BaseT (UTP)	CSMA/CD (Half-Duplex)
Fast Ethernet	100 Mbps	100BaseTX (Cat 5 UTP), 100BaseFX	CSMA/CD / Switched Full-Duplex
Gigabit Ethernet	1 Gbps	1000BaseT (Cat 5e UTP), 1000BaseLX	Switched Full-Duplex
IEEE 802.11 (Wi-Fi)	Up to 9.6 Gbps (Wi-Fi 6)	2.4 GHz / 5 GHz / 6 GHz Radio	CSMA/CA with RTS/CTS Handshake

Topic 42 & 43: Internet Protocol (IPv4 Header & Subnetting) [UPLOADED PYQ]

IPv4 Header Field [UPLOADED PYQ]	Bit Width	Functional Operational Purpose
Version	4 bits	Identifies IP version ('0100' for IPv4).
Header Length (IHL)	4 bits	Header length in 32-bit words (Minimum value = 5 $\implies$ 20 bytes).
Total Length	16 bits	Total datagram length (Header + Payload) in bytes (Max = 65,535 bytes).
Identification, Flags, Offset	16 + 3 + 13 bits	Handles IP fragmentation. 'DF' = Don't Fragment, 'MF' = More Fragments. Offset in 8-byte units.
Time to Live (TTL)	8 bits	Decrement at each router hop. When TTL = 0, packet is dropped (prevents infinite loops).
Protocol	8 bits	Demultiplexes to Transport layer ('6' for TCP, '17' for UDP, '1' for ICMP).
Header Checksum	16 bits	1's complement sum of header fields recalculated at each router hop.

Subnetting Formulations: - Number of Created Subnets = $2^{\text{borrowed bits}}$ - Total IP Addresses per Subnet = $2^{\text{host bits}}$ - Usable Host Addresses per Subnet = $2^{\text{host bits}} - 2$ (subtracting Subnet Network ID and Directed Broadcast Address).

Topic 44: Transport Protocols (TCP vs. UDP) [UPLOADED PYQ]

Feature	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection Model	Connection-Oriented (3-Way Handshake: SYN $\rightarrow$ SYN-ACK $\rightarrow$ ACK).	Connectionless (no handshake; sends datagrams immediately).
Reliability	100% Reliable (Sequence numbers, ACKs, retransmissions on loss).	Unreliable Best-Effort (no ACKs, lost packets are not retransmitted).
Ordering & Streaming	Guaranteed strictly in-order byte stream reassembly.	No ordering guarantee; datagram boundary preserved.
Flow & Congestion Control	Full sliding window flow control + AIMD Congestion Control.	Zero flow control, zero congestion control.
Typical Applications	Web (HTTP/HTTPS), Email (SMTP), File Transfer (FTP), SSH.	DNS queries, Video streaming (RTP), VoIP, DHCP, Online Gaming.

Topic 45 – 48: Routing Protocols (Distance Vector vs. Link State vs. Path Vector)

Routing Protocol	Mathematical Algorithm	Scope & Protocol	Key Characteristics
Distance Vector	Bellman-Ford: $D_x(y) = \min_v \{c(x, v) + D_v(y)\}$	Intra-AS (RIP)	Exchanges vector with neighbors; susceptible to Count-to-Infinity.
Link State	Dijkstra's Shortest Path Tree Algorithm	Intra-AS (OSPF)	Floods Link State Advertisements (LSA) globally; builds complete topological map.
Path Vector	Policy-based sequence of Autonomous System (AS) hops	Inter-AS (BGP-4)	Prevents loops by checking if local AS number is present in path vector.

Topic 49 & 50: Congestion Control & Traffic Shaping (Leaky vs. Token Bucket)

Traffic Shaper	Operating Mechanism	Burst Handling Capability
1. Leaky Bucket	Input packet bursts fill a buffer with fixed capacity. Water (packets) leaks out at a strictly constant uniform output rate . Excess bursts overflow and are discarded.	Completely eliminates bursts; outputs strictly smoothed constant bit-rate traffic.
2. Token Bucket	Tokens generate into the bucket at a constant rate r up to capacity $\bar{b}$. A packet can only transmit if it captures tokens.	Allows controlled, regulated bursts up to bucket capacity $\bar{b}$ while capping average rate at r .

Topic 51 – 54: Application Layer Services (SMTP, DNS, HTTP, DHCP)

🧠 Summary of Core Application Protocols & Ports

- **SMTP (Port 25/587)**: Simple Mail Transfer Protocol; pushes email between client and server and between mail relays.
- **DNS (Port 53 – UDP/TCP)**: Hierarchical Domain Name System resolving domain names to IP addresses (Records: 'A' for IPv4, 'AAAA' for IPv6, 'MX' for Mail Exchange, 'CNAME' for Canonical Alias).
- **HTTP/HTTPS (Port 80/443)**: Stateless request-response web protocol. Status codes: '200 OK', '301 Moved Permanently', '404 Not Found', '500 Internal Server Error'.
- **DHCP (Port 67/68 – UDP)**: Dynamic Host Configuration Protocol assigning IP addresses dynamically via the **D-O-R-A** sequence (Discover → Offer → Request → Acknowledge).

🌐 [UPLOADED PYQ] Complete Life Cycle: What Happens When You Type 'https://example.com'?

1. **Host IP Configuration (DHCP)**: Client uses DHCP DORA sequence to acquire local IP, Subnet Mask, Default Gateway, and DNS Server.
2. **DNS Resolution**: Browser queries local DNS cache, then queries Recursive Resolver → Root Server → TLD Server (`.com`) → Authoritative Server to obtain target IP for `example.com`.
3. **ARP Translation**: Host uses Address Resolution Protocol (ARP) to map Default Gateway IP to its physical 48-bit MAC address.
4. **TCP 3-Way Handshake**: Client sends `SYN` (Port 443) → Server replies `SYN-ACK` → Client confirms `ACK` (Connection Established).
5. **TLS 1.3 Handshake**: Client and server negotiate cipher suites, authenticate server X.509 certificate, and exchange ephemeral Diffie-Hellman keys to establish encrypted session.
6. **HTTP GET Request & IP Routing**: Browser transmits encrypted `GET / HTTP/2` request. Packet is encapsulated with IP header and routed across intermediate Autonomous Systems via OSPF and BGP-4.
7. **Server Processing & Response**: Web server processes request and returns HTTP `200 OK` response with HTML/CSS payload.
8. **Browser Rendering**: Browser parses HTML DOM tree, fetches CSS/JS, and renders the visual webpage!

🧠 M5 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ] Explain the structure and working of the IPv4 Header with the purpose of all major fields. (10 Marks)

The IPv4 header has a minimum size of 20 bytes (up to 60 bytes with options):

1. *Version (4b)*: '4' for IPv4.
2. *IHL (4b)*: Header length in 32-bit words (min 5).
3. *Total Length (16b)*: Header + Data in bytes.
4. *Identification (16b)*, *Flags (3b)*, *Fragment Offset (13b)*: Fragmentation management for MTU limits.
5. *TTL (8b)*: Hop limit decremented at each router to prevent packet looping.
6. *Protocol (8b)*: Transport demultiplexing ('6' = TCP, '17' = UDP).
7. *Header Checksum (16b)*: Error detection across header fields.
8. *Source IP (32b) & Destination IP (32b)*: Logical sender and receiver addresses.